3. One difference between force and matter fields is expressed by Wolfgang Pauli's *exclusion principle.* This principle shows that whereas a huge number of force particles (like photons) can combine to produce fields accessible to a prequantum physicist such as Maxwell, fields that you see every time you enter a dark room and turn on a light, matter particles are generally excluded by the laws of quantum physics from cooperating in such a coherent, organized manner. (More precisely, two particles of the same species, such as two electrons, are excluded from occupying the same state, whereas there is no such restriction for photons. Thus, matter fields do not generally have a macroscopic, classical-like manifestation.)

4. In the framework of quantum field theory, every known particle is viewed as an excitation of an underlying field associated with the species of which that particle is a member. Photons are excitations of the photon field—that is, the electromagnetic field; an up-quark is an excitation of the up-quark field; an electron is an excitation of the electron field, and so on. In this way, all matter and all forces are described in a uniform quantum mechanical language. A key problem is that it has proved very difficult to describe all the quantum features of gravity in this language, an issue we will discuss in Chapter 12.

5. Although the Higgs field is named after Peter Higgs, a number of other physicists—Thomas Kibble, Philip Anderson, R. Brout, and François Englert, among others— played a vital part in its introduction into physics and its theoretical development.

6. Bear in mind that the field's *value* is given by its distance from the bowl's center, so even though the field has zero *energy* when its value is in the bowl's valley (since the height above the valley denotes the field's energy), its value is not zero.

7. In the text's description, the value of the Higgs field is given by its distance from the bowl's center, and so you may be wondering how points on the bowl's circular valley— which are all the same distance from the bowl's center—give rise to any but the *same* Higgs value. The answer, for the mathematically inclined reader, is that different points in the valley represent Higgs field values with the same magnitude but different phases (the Higgs field value is a complex number).